

BHOOMISYNC

# DIGITAL LAND SURVEY REPORT

AI-Powered Cadastral Intelligence & Drone Resurvey Dossier

STATUS: AI DETECTED

|                     |                                                         |
|---------------------|---------------------------------------------------------|
| Report Number:      | BHOOMI/RJ/UDP/2026/010002-V1                            |
| Survey Mission ID:  | SUR-2026-001                                            |
| Parcel / Khasra No: | Khasra #102                                             |
| Village / Tehsil:   | Haripura, Tehsil Girwa                                  |
| District & State:   | Udaipur, Rajasthan                                      |
| Survey Date:        | 2026-09-01                                              |
| Coordinate System:  | EPSG:4326 (WGS84) / Projected UTM Zone 43N (EPSG:32643) |
| Data Access Role:   | SURVEYOR                                                |

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# 1. Executive Summary

This digital survey report provides an authoritative spatial synthesis of drone aerial photogrammetry, airborne LiDAR elevation profiling, and RTK GNSS ground truth positioning compared against historical and official village revenue records. The findings summarize land-use classification, precision surface area, multi-boundary spatial alignment, and surveyor verification status.

| Metric Parameter       | Official Revenue Record | Drone Sensor Survey | Verified Boundary | Variance / Status |
|------------------------|-------------------------|---------------------|-------------------|-------------------|
| Cadastral Area         | 9,800.00 m²             | 9,800.00 m²         | 9,800.00 m²       | +0.00%            |
| 3D Surface Area        | Not in Record           | 9,976.40 m²         | 9,976.40 m²       | +1.8% vs Planar   |
| Field Perimeter        | 445.00 m                | 448.60 m            | 448.60 m          | +3.60 m           |
| Primary Land Use       | AGRICULTURAL            | AGRICULTURAL CROP   | AGRICULTURAL CROP | AI Conf: 94.7%    |
| Boundary Match (IoU)   | Baseline (1.00)         | IoU: 0.974          | Sign-off Complete | WITHIN TOLERANCE  |
| Potential Encroachment | N/A                     | None Detected       | Verified          | CLEAN TITLE       |

## Key Survey Observations:

- True geodesic measurement on WGS84 ellipsoid confirms parcel area matches official khasra within 0.20% tolerance.
- RTK GNSS fix quality maintained centimeter-level positioning (Horizontal  $\pm 1.4$  cm, Vertical  $\pm 2.1$  cm).
- Deep-learning aerial classification identified 78% active agricultural crop and 12% seasonal fallow.
- Digital boundary coordinates cross-referenced with 2021 historical cadastre showing zero structural encroachments.

## 2. Parcel Information & Ownership Title

### Cadastral Record Details:

|                   |                                         |                    |                 |
|-------------------|-----------------------------------------|--------------------|-----------------|
| Khasra Number     | 102                                     | Village            | Haripura        |
| Subdivision       | 1                                       | Tehsil             | Girwa           |
| District          | Udaipur                                 | State              | Rajasthan       |
| Centroid Latitude | 24.584000° N                            | Centroid Longitude | 73.713200° E    |
| Tenure Category   | KHATEDARI (Permanent Agricultural Land) | Tax Status         | PAID_UP_TO_DATE |

### Registered Landowners (RBAC Filtered):

| # | Landowner Name     | Owner Type | Share % | Contact Ref       | Verification ID |
|---|--------------------|------------|---------|-------------------|-----------------|
| 1 | Suresh Kumar Dangi | INDIVIDUAL | 100.0%  | HASH-TEL-91290002 | OWN-HR-002      |

### 3. 2D GIS Cadastral Map & Boundary Overlays

Visual representation of multi-source boundary layers: Official Revenue Cadastre (Blue), Historical Baseline (Purple), Drone AI Detected Bunds (Yellow), and Surveyor Verified Boundary (Green).



## 4. Precision Geospatial & Terrain Analysis

| Measurement Property    | Metric Value (m²) | Hectares (ha) | Acres (ac) | Calculation Engine          |
|-------------------------|-------------------|---------------|------------|-----------------------------|
| Planar Surface Area     | 9,800.00 m²       | 0.9800 ha     | 2.4216 ac  | WGS84 Ellipsoidal Integral  |
| 3D Terrain True Surface | 9,976.40 m²       | 0.9976 ha     | 2.4652 ac  | LiDAR DEM Triangulation     |
| Official Record Area    | 9,800.00 m²       | 0.9800 ha     | 2.4216 ac  | Revenue Record (Jamabandi)  |
| Surveyor Verified Area  | 9,800.00 m²       | 0.9800 ha     | 2.4216 ac  | GCP-Corrected Final Polygon |

### Elevation & Slope Profile:

|                   |               |                    |                                             |
|-------------------|---------------|--------------------|---------------------------------------------|
| Perimeter Length  | 448.60 metres | Mean Terrain Slope | 4.2° (Gentle Agricultural Gradient)         |
| Minimum Elevation | 582.40 m MSL  | Maximum Elevation  | 594.10 m MSL                                |
| Elevation Relief  | 11.70 metres  | Drainage Direction | South-South-West towards Haripura Reservoir |

## 5. Drone Survey Flight & Sensor Telemetry

| Sensor & Parameter                   | Configuration & Technical Value                     | Quality Assurance Benchmark                |
|--------------------------------------|-----------------------------------------------------|--------------------------------------------|
| RGB Aerial Camera                    | Sony RX0 II Ultra-Compact Survey Camera (DSC-RX0M2) | Full-frame Mechanical Shutter              |
| Image Ground Sampling Distance (GSD) | 1.25 cm/pixel                                       | Exceeds Survey of India <5cm standard      |
| Total Raw Aerial Frames              | 240 geotagged images                                | 80% Forward / 75% Lateral Overlap          |
| Airborne LiDAR Scanner               | Livox Mid-360 Solid-State 3D LiDAR (Mid-360)        | Class 1 Eye-Safe Dual-Return 905nm         |
| LiDAR Point Cloud Total              | 14,500,000 points                                   | Classified Ground / Vegetation / Structure |
| Point Cloud Density                  | 185.4 pts/m²                                        | High-density terrain penetration           |
| RTK/GNSS Fix Status                  | FIXED                                               | Multi-frequency L1/L2/L5 RTK Fixed Fix     |
| Mean Horizontal Accuracy             | ±1.2 cm                                             | Sub-2cm Cadastral Accuracy Target          |
| Mean Vertical Accuracy               | ±2.1 cm                                             | Sub-3cm Topographical Target               |
| Flight Altitude (AGL)                | 60.0 metres                                         | Optimized terrain-following trajectory     |

## 6. AI Land-Use & Land-Cover (LULC) Classification

Primary Land Classification: AGRICULTURAL CROP (Overall AI Confidence: 94.7%)

| Land-Use Category   | Class Description               | Coverage Area (m²) | Share (%) | AI Model Confidence     |
|---------------------|---------------------------------|--------------------|-----------|-------------------------|
| AGRICULTURAL_CROP   | Cultivated Crop (Wheat/Mustard) | 7,644.00 m²        | 78.0%     | HIGH (95%+) (96.0%)     |
| AGRICULTURAL_FALLOW | Seasonal Fallow                 | 1,176.00 m²        | 12.0%     | MEDIUM (90-95%) (92.0%) |
| VEGETATION_TREES    | Field Boundary Trees & Shrubs   | 588.00 m²          | 6.0%      | LOW (89.0%)             |
| WATER_HARVESTING    | Irrigation Bund / Channel       | 392.00 m²          | 4.0%      | MEDIUM (90-95%) (94.0%) |

**MANDATORY AI NOTICE:** AI-generated classification — predictions are generated via deep-learning computer vision models and are subject to field verification by an authorized cadastral surveyor.

## 7. Multi-Boundary Alignment & Cadastral Comparison

| Spatial Alignment Parameter   | Observed Variance             | Tolerance Threshold | Status Compliance      |
|-------------------------------|-------------------------------|---------------------|------------------------|
| Intersection-over-Union (IoU) | 0.974                         | $\geq 0.950$        | PASSED (High Overlap)  |
| Maximum Vertex Displacement   | 0.42 m                        | $\leq 0.75$ m       | PASSED (Within Bounds) |
| Mean Boundary Displacement    | 0.18 m                        | $\leq 0.30$ m       | PASSED (Sub-20cm RMS)  |
| Centroid Spatial Drift        | 0.22 m                        | $\leq 0.50$ m       | PASSED (No Shift)      |
| Area Discrepancy              | +0.00 m <sup>2</sup> (+0.00%) | $\pm 1.0\%$         | PASSED (Consistent)    |
| Perimeter Variation           | +3.60 m                       | $\pm 2.0\%$         | PASSED (Consistent)    |



## 8. Historical Change Detection & Temporal Cadastre

| Epoch / Year     | Source of Cadastral Baseline          | Recorded Area            | Observed Morphological Changes                     |
|------------------|---------------------------------------|--------------------------|----------------------------------------------------|
| 2015 Cadastre    | Revenue Settlement Survey (Paper Map) | 12,450.00 m <sup>2</sup> | Initial traditional earthen bund demarcation.      |
| 2021 Drone Pilot | Optical Aerial Survey 5cm GSD         | 12,460.00 m <sup>2</sup> | South boundary bund reinforced with stone markers. |
| 2026 Resurvey    | BhoomiSync RTK + LiDAR Fusion         | 9,800.00 m <sup>2</sup>  | Modern high-precision geodesic boundary baseline.  |

## 9. Encroachment Risk Assessment

|                         |                                                                      |
|-------------------------|----------------------------------------------------------------------|
| Assessment Parameter    | Analysis Finding & Details                                           |
| Encroachment Status     | NO POTENTIAL ENCROACHMENT DETECTED                                   |
| Alert Reference ID      | NONE (Clean Field Geometry)                                          |
| Discrepancy Severity    | NONE                                                                 |
| Estimated Affected Area | 0.00 m²                                                              |
| Spatial Location Note   | Field perimeter is strictly consistent with village revenue records. |

**LEGAL NOTICE: POTENTIAL ENCROACHMENT NOTICE:** Encroachment markers and boundary anomalies identified herein represent preliminary spatial discrepancies between cadastral records and aerial sensor datasets. They constitute 'Potential Encroachments' and require formal ground demarcation and field verification.

## 10. Surveyor Verification & Digital Audit Trail

|                          |                                                 |                         |                     |
|--------------------------|-------------------------------------------------|-------------------------|---------------------|
| Field Verification State | AI_DETECTED                                     | Verification Date       | 2026-09-01          |
| Cadastral Surveyor       | Vikram Singh (Cadastral Surveyor, GCP Verified) | Surveyor ID / License   | RAJ-REV-CAD-4412    |
| Digital Certificate Hash | SIG-SHA256-0824924C8A5623B8                     | Snapshot SHA-256 Anchor | e618779e0167b4b2... |

### Surveyor Operational Remarks:

"Aerial boundary verified with RTK Ground Control Points. Alignment adheres to Rajasthan Land Revenue Act standards."

|                                   |                                    |                               |
|-----------------------------------|------------------------------------|-------------------------------|
| Prepared By:                      | Reviewed By:                       | Approved By:                  |
| Cadastral Surveyor (GCP Verified) | Assistant Settlement Officer (ASO) | Tehsildar / Revenue Authority |

**End of Official Report.** Generated by BhoomiSync Intelligent Cadastral Platform.